Secret Codebreakers & The Enigma Machine

Key Enquiry Question: How did Alan Turing and the secret minds at Bletchley Park crack the unbreakable German cipher?

LESSON 08
KEY STAGE 2



Source A: *An Enigma cipher machine showing its alphabetical keyboard, illuminating lampboard, and spinning rotor wheels. (Bletchley Park Trust)*

HISTORICAL READING • WHAT THIS TELLS US

The German military used an electrical cipher typewriter called the **Enigma machine** to scramble battle orders. With spinning rotor wheels and plugboard cables, Enigma could scramble a message into **158 million million million combinations**.

At secret Bletchley Park in Buckinghamshire, genius mathematician Alan Turing and thousands of young codebreakers designed electromechanical "Bombe" machines that tested combinations at lightning speed, breaking the code daily.

Task 1: Artifact Detective • Alan Turing's Bombe Machine

PRIMARY EVIDENCE



Source B: *The towering electromechanical Bombe machine in Hut 11, with dozens of spinning coloured drums designed to decipher Enigma keys.*

1. Look closely at Source B. Why did Alan Turing realize that only an electromechanical machine (the Bombe) could test thousands of cipher combinations fast enough to save lives?

2. Looking at Source A and Source B, why was the midnight reset of German cipher keys such a desperate daily race against time for codebreakers?

Task 2: Active Chronological Challenge • The Codebreaking Pipeline

CHRONOLOGICAL UNDERSTANDING

STAGE 1

Radio Intercept (Y-Stations)

Coastal listening stations intercept Morse code radio signals transmitted by German U-boats and armies.

Why did listening operators need lightning-fast ears to transcribe Morse code dits and dahs?

STAGE 2

Hut 8 & The Turing Bombes

Coded slips rushed to Bletchley. Bombes whirr through rotor settings to find the day's secret key.

What went through codebreakers' minds when the machine suddenly stopped on a winning match?

STAGE 3

'Ultra' Secret Intelligence

Decrypted battle orders delivered to Churchill in locked yellow dispatch boxes to re-route convoys.

Why was keeping the secret that Britain had cracked Enigma just as critical as breaking the code?

Task 3: Visual Investigation • Draw an Enigma Cipher Wheel & Decoding Mechanisms

TACTILE INVESTIGATION

Draw your cipher wheel showing scrambled alphabet letters, rotor contacts, and electrical connections.

1. The Plugboard (Steckerboard)
How did swapping 10 pairs of letters before the rotors turned create millions of extra combinations?

2. The Three Spinning Rotors
Why did the rotor wheel stepping forward with every keystroke prevent simple letter-frequency decoding?

3. The Reflector (Umkehrwalze)
Why was the rule that a letter could never be encrypted as itself the fatal flaw Turing exploited?

4. The 30-Year Secrecy Oath
Why did over 10,000 workers at Bletchley never tell their spouses or families what they did for 30 years?

Task 4: "Thinking Heads" • Contrasting Historical Perspectives

CONTRASTING PERSPECTIVES

1 Alan Turing (Mathematician & Pioneer)

What mathematical breakthrough convinced you that logic machines could defeat human ciphers and save millions of Allied lives?

2 Wren Operating the Bombe in Hut 11

What does it feel like to spend 8 hours in deafening mechanical rotor noise knowing your dials might save a convoy at sea tonight?

Task 5: Mission Challenge • Shortening the War

HISTORIAN CHALLENGE

Big Question: Historians calculate that Bletchley Park shortened the war by up to two years. How did silent codebreaking in secret wooden huts save millions of civilian and soldier lives?

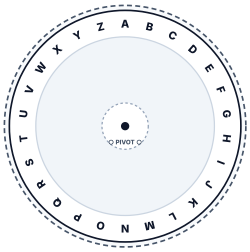
BLETCHLEY PARK CONCENTRIC ENIGMA CIPHER WHEEL & DECODER DISPATCH

Name: _____

✂ **Hands-On Primary Artefact Activity:** Cut out both circular cipher wheels below. Place the smaller Inner Scrambler Disc over the Outer Ring and pierce a split-pin (paper fastener) through the center pivot dot. Rotate the inner wheel to decode the secret wartime naval intercepts!

✂ CUT OUT BOTH WHEELS

1. OUTER BASE RING (PLAINTEXT ALPHABET)



2. INNER ROTATING SCRAMBLER DISC (CIPHER)



Assembly Tip: Cut out both wheels. Place the small wheel directly on top of the big wheel. Push a split-pin through the center dots and fold flat.

GOVERNMENT CODE & CYPHER SCHOOL

BLETCHLEY PARK • HUT 6 NAVAL INTERCEPT SECTION

MOST SECRET • ULTRA

INTERCEPT DISPATCH #1 • ATLANTIC CORRIDOR

KEY: Shift +3 (Align Arrow to 'D')

Enemy U-boat radio message intercepted at 02:40 hours. Find each cipher letter on the Inner Wheel, and read the real letter on the Outer Wheel:

`FRQYRB VDIH LQ SRUW`

DECODED: _____

INTERCEPT DISPATCH #2 • SECRET BOMBE ALERT

KEY: Shift +3 (Align Arrow to 'D')

Urgent signal decoded from Hut 8 using Alan Turing's electromechanical Bombe machine:

`WKH HQJPD FRGH LV EURNHQ`

DECODED: _____

MISSION CHALLENGE: ENCODE A MESSAGE FOR A PARTNER

MY SECRET KEY: [ ]

Choose your own secret key letter. Write a 3-word secret Allied message in code. Swap exercise books with your classmate to see if they can crack your code!

YOUR CIPHER CODE: _____

PARTNER'S DECODE: _____

Codebreaker in Charge: _____

Official Secrets Act 1939 • Station X Commendation

PASSED HUT 6 VERIFICATION